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METHODS AND SYSTEMS FOR POLISHING AND RECOVERING ALUMINUM FROM A WASTE MATERIAL

Abstract

A method includes one or more of the following five steps: rough processing, comminuting the material, washing the material with acid, and collecting/sorting the material to recover an aluminum product or a very pure aluminum product. A system may execute one or more of these steps to recover an aluminum product or a very pure aluminum product.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/776,605, filed Jul. 18, 2024, which is a continuation of U.S. patent application Ser. No. 16/465,131, filed May 29, 2019, which is a national stage application, filed under 35 U.S.C. § 371, of International Patent Application No. PCT/US2017/063812, filed on Nov. 29, 2017, which claims the benefit of U.S. Provisional Patent Application Nos. 62/427,793, filed Nov. 29, 2016, and 62/506,329 filed on May 15, 2017, each of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This application relates to systems and methods for separating, purifying, or polishing aluminum from ash zorba or zorba. More particularly, this application relates to systems and methods for polishing aluminum (e.g., silicon, glass, cast aluminum, copper, brass, zinc, lead, and any other impurities) from IBA zorba or zorba using a wet process. This application also relates to systems and methods for purifying or upgrading aluminum recovered from IBA or zorba by reducing impurities and sealing the metals from further oxidation.

BACKGROUND

[0003] Recycling of waste materials is highly desirable from many viewpoints, not the least of which are financial and ecological. Properly sorted recyclable materials can often be sold for significant revenue. Many of the more valuable recyclable materials do not biodegrade within a short period, and so their recycling significantly reduces the strain on local landfills.

[0004] However, in many instances there has been no cost-effective method for achieving the necessary sorting or for recovering a highly pure aluminum product. While aluminum recycling has been available for some time, this technique is plainly ineffective for recovering a high purity aluminum product from zorba, whether from ash, automobile waste residue or other waste material.

[0005] As a result, there is a need for a cost-effective, efficient method and system for recovering purified aluminum from zorba.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] This disclosure is illustrated in the figures of the accompanying drawings which are meant to be illustrative and not limiting, in which like references are intended to refer to like or corresponding parts, and in which:

[0007] FIG. 1 illustrates an exemplary flow diagram for recovering metals from a waste stream according to the present disclosure; and

[0008] FIG. 2 illustrates an exemplary system for recovering metals from a waste stream according to the present disclosure.

SUMMARY

[0009] One aspect of this application includes a method for separating waste material that includes the steps of providing waste material having aluminum and heavy metals; rough processing the waste material to remove the heavy metals; floating the material to remove additional impurities;

sizing the material to predetermined sizes; comminuting the material, wet screening the material, washing the material with an acid produce darkened material; and sorting the material to separate the darkened material and a purified aluminum material. The purified aluminum material may be further purified using an eddy current.

[0010] Another aspect of this application includes a system for preparing an aluminum product that has a source of waste material having aluminum and heavy metals; a rough processing device capable of removing the heavy metals from the waste material; a sizing device that sizes the waste material to predetermined sizes; a comminution device that is a ball mill; a first screen that is a two-stage screen; an acid wash; at least one optical sorter; and a collector for the aluminum product.

DETAILED DESCRIPTION

[0011] Embodiments of methods and systems for the separation and recovery of metal, particularly aluminum, from a waste material using comminution and acid washes are disclosed. The disclosed embodiments are particularly well-suited for producing a high-grade aluminum product. Some embodiments of the disclosed method include one or more of the following five steps: (1) rough processing, (2) comminuting the material, (4) washing the material with acid, and (5) collecting/sorting the material to recover an aluminum product or a very pure aluminum product.

[0012] This application includes methods and systems for improving the quality and aesthetics of non-ferrous metals, particularly aluminum, through the use of comminution (e.g., a ball mill). Specific embodiments may be carried out in wet processes. That is, the wet process can include a slurry of incinerator bottom ash (IBA), ash, or automobile shredder residue zorba or zorba that has aluminum content. In illustrative example, the product can be a substitute for prime grade aluminum suitable for high grade application, e.g., coil, strip, sheet, plates, tube, bar and structural products (e.g., prime alloys Series 2000, 3000, 5000, etc.). In one example, the product is or has the specifications of Aluminum 3105, which has an aluminum purity of about 98% (e.g., 98% Aluminum, 0.6 silicon, 0.4 Zinc, and other elements). In another example, the product is or has the specifications of wrought aluminum alloy predominantly 3105 aluminum alloy or Grade 1 DEOX aluminum.

[0013] IBA is the ash that is left over after waste is burnt in an incinerator. Municipal energy from waste plants that use incineration burn a wide range of municipal wastes and therefore the term 'ash' is slightly misleading because it is not all powdery but contains glass, brick, rubble, sand, grit, metal, stone, concrete, ceramics and fused clinker. Examples of bottom ash for use with certain embodiments include the residues which are obtained when refuse of all kinds such as city garbage, sewer sludge and industrial waste is incinerated inside an incinerator. Examples of ash under consideration in this application include substances which fly off with the discharge gas when such refuse as described above is incinerated in an incinerator and is collected in its discharge gas processing system, or substances collected mainly by a dust collector provided in such a discharge gas processing system. In some examples, ash having an aluminum content of greater than 70% weight percentage is particularly useful.

[0014] In illustrative embodiments, the material may be a zorba. Institute of Scrap Recycling Industries (ISRI) defines zorba as "shredded mixed nonferrous metals consisting primarily of aluminum generated by eddy current separator or other segregation techniques." Other nonferrous metals found in zorba include copper, lead, brass, magnesium, stainless steel, nickel, tin and zinc in elemental or alloyed (solid) form.

[0015] In one embodiment, the incinerator bottom ash has undergone rough processing, which may be accomplished by identifying a physical property that is more pronounced in the metal than in the mineral matrix material and using that property for separation. In one example, the waste material may be rough processed using an eddy current, which results in light material (typically referred to as zorba, incinerated zorba or incinerated non-ferrous). Although not shown separately, a portion of bottom ash with water content of 40 weight % can be preliminarily dried to reduce its water content

to 10 weight % or less and then subjected to a sifting process and a crushing process, to create particles that could pass through a screen with a mesh size of 100 mm. Generally, ash taken out of an incinerator has a substantially large water content of about 20-50 weight % because of a moistening process, as explained above. If the water content exceeds 10-15 weight %, the adhesive force of bottom ash increases sharply and its angle of repose also increases rapidly such that the ash becomes likely to get stuck to a hopper and the conveyor, clogging various openings.

[0016] FIG. 1 shows a basic flow diagram according to a first embodiment. The IB A zorba or zorba **10** is processed using rough processing step to remove ferrous metals, heavy metals and other materials. Properties typically employed for the extraction of metals from zorba include, but are not limited to magnetic susceptibility, electrical conductivity, and density. For example, the IB A zorba or zorba **10** can be processed directly through “rough processing” using density separation **20**, which is at about or at least 3.5 S.G. A density separation **20** at about 3.5 S.G. allows for the removal or recovery of heavy metals or “heavies.” The material may also be subject to flotation **30**, which removes impurities and other materials. Other density specifications may be used with rough processing.

[0017] In one embodiment, the particles are sized so as to optimize sorting, e.g., through an optical sorter or another type of sorter. The materials can be segregated into discrete size ranges based on, e.g., commercially available equipment and specifications. Sizing particles at step **40** may, in certain embodiments, include several steps of sizing particles. In one example, step **40** may include sizing the slurry with one or more screens that sizes the material into four segmented portions of particles having different respective sizes. In one embodiment, the four segmented portions of particles include (1) a segmented portion of particles having a dimension less than the predetermined size, which is about 2.0 mm in major dimension, (2) a segmented portion of particles having a major dimension between about 2.0 mm and 6 mm, (3) a segmented portion of particles having a major dimension between about 6.0 mm and 18 mm, and (3) a segmented portion of particles having a major dimension greater than 18 mm or between 18 mm and about 40 mm. In other embodiments, each of the segmented portions of particles may be defined by alternate size ranges, depending upon the characteristics of the screen(s) used at step **40**. After step **40**, the segmented portion of particles having the dimension less than the predetermined size are provided for processing in later steps.

[0018] The sized materials can be transported and fed to a comminution system/scrubbing device, e.g., a ball mill **50**. The materials pass to or are fed to a ball mill, which pulverizes and flattens the material. The ball mill can pulverize non-metallic attachments such as glass, slag and rocks, e.g., that can be entrapped with the aluminum. Optionally, the metals that are flattened (because of their malleability properties) to increase the surface area and enhance the separation process.

[0019] After the scrubbing device or ball mill **50** step, the material can be subjected to a wet screening step **60**. Wet screening units are designed to efficiently screen out fines and classify oversize particles.

[0020] At the etching step **70** and after the screening process, the material is subjected to or etched by an acid bath or wash. The acid suppresses the oxidation of the metals and improves the aesthetics of the material or product. Further, the acid blackens non-aluminum materials (e.g., cast aluminum, magnesium, and zinc). One exemplary acid includes 40% sulfuric acid—this acid can be sprayed from a sump holding the acid on a screen in a hood having a neutralizing base (e.g., caustic soda or sodium hydroxide). The sulfuric acid can then drain from a sump back to a holding container/tote for recirculation. A scrubber can be bled from its bottom to provide a rinse water as well. The pH may be controlled in both the sump and the scrubber. Other acids, e.g., hydrochloric acid can be used. In one example, the pH of the acid is between 1.0 and 4.0. In one example, the acid can be applied through a screw.

[0021] In one exemplary embodiment, the acid wash/bath darkens, blackens and/or alters the color of certain materials. For example, zinc, magnesium, and cast aluminum, can be darkened or

blackened by the acid wash or bath. Certain aluminum cast and wrought materials can be blackened or darkened by the acid bath or wash. Copper and other heavy metals retained their color. The aluminum retains its relatively grayish color after the acid wash or bath.

[0022] After the etching step **70**, the material may be further screened, e.g., at another wet screening step **80**. At this step **80** and in one example, the materials screen for fines, which are less than about 6 mm or less than 1 mm. The materials less than 1 mm typically are copper and other impurities. The acid can be removed during this step **80**.

[0023] After the materials have been etched, the material may be subjected to one or more cascade of sorters **90**, **100**, **110**. In one exemplary embodiment, the acid washed material or the material can be subjected to optical sorters, stainless steel sorters, infrared sorters, camera sorting machines, and the like. Such sorters can be used to separate different types of the darkened materials from the grayer aluminum. In other words, the darkened materials can be removed from the loose gray aluminum, which results in a clean aluminum product. In one example, the sizing of the material correlates with the pixel limits of the sorter.

[0024] The sorter may be selected to remove the darkened and blackened material. In one example, the sorter **90** was an optical sorter screening for copper and brass based on color.

[0025] The material from the sorters **90**, **100**, **110** may be further treated with eddy currents **120**, density separation, and other devices. For example, the use of eddy current **120** allows for additional organic material and other impurities to be removed from the material.

[0026] In one exemplary embodiment, the dewatered materials or the material are screened at about 1 mm and about 6 mm. Materials greater than 6 mm can be reprocessed through the ball mill, and materials less than 1 mm can be processed using other methods and techniques. The materials less than 1 mm typically are copper and other impurities. The materials between 1 mm and 6 mm are washed and/or dewatered. The overall metal content/purity increases because the pulverized non-metallics are washed through the de-watering screen. There can be a hood over the screen to allow the materials to be washed or sprayed.

[0027] FIG. **2** shows an exemplary system for preparing an aluminum product and capable of executing steps of exemplary methods. As can be seen, the system **200** includes a source of waste material having aluminum and heavy metals or feeder **210**; a rough processing device capable of removing the heavy metals from the waste material; a sizing device that sizes the waste material to predetermined sizes; a comminution device that is a ball mill **220**; a first screen that is, e.g., a two-stage screen **230**; an acid wash (e.g., an screw with an acid bath) or screw **240**; a wet screen **250**, at least one optical sorter **260**, **270**, **271**, **280**; and a collector for the aluminum product. The system may have a eddy current separator **290**, and vision bridge **215**, and water supply W. This system may operate according to the layouts shown in FIG. **2**. Sizing or discrete sizing can improve separation of materials.

[0028] In this embodiment, the ball mill is a type of pulverizer, which is a cylindrical device used in grinding and pulverizing materials. Ball mills rotate around a horizontal axis, partially filled with the material to be ground plus a grinding medium. Different materials are used as media, including ceramic balls, flint pebbles and stainless steel balls. In one example, the balls have a radius between 4-5 mm. In one example, the ball mill is a rotary pulverizer having a plurality of liner sections on its inner periphery that are made of a composite material. The innermost layer or base of the liner section can be made from a resilient rubber material formed with an undulating outer surface.

[0029] One consequence of the ball mill processing is degradation and pulverization of glass that is bonded or adhered to the aluminum in the ash. For example, the ball mill acts to break, crush and separate any glass or silica mixed with the alumina in the ash.

[0030] In the separation process in accordance with the present embodiment, copper (e.g., tubes and wires) are pulverized and flattened into denser structures, which then sink in water, whereas the aluminum component is flattened, which increases floatation properties. The aluminum, glass and

copper can be efficiently separated, e.g., by an ordinary screening operation. The aluminum, glass and copper can also be efficiently separated into their components, e.g., by specific gravity separation, such as centrifugal, inertial, and gravity classifications.

[0031] An optional dewatering step can be used to remove water and reduce the moisture in the material. This material can contain between 25% and 50% moisture by weight. The material can be placed in dryers where the moisture content is brought down to less than 20% by weight. The dryers operate at temperatures of greater than 300 degrees F., and the temperature can depend on the metal content and the desired level of moisture for the recovered product. The resulting volume of recovered metal powder is reduced over the previous dewatering step by as much as 30% to 50% by volume. The drying process drives off the moisture and other compounds that are not metallic leaving the metals in the resulting dry materials heavily concentrated.

[0032] In general, this disclosure provides an economical method of processing a bottom ash and fly ash using a ball mill. The silica content of the product is surprisingly low and allows for aluminum alloy sheet applications. The crushing and pulverizing of the silica allows for its removal. In certain examples, the weight percent of aluminum increased from about 92% to 97% and in another example, the weight percent of aluminum increased from 86% to about 99%. In many cases, the material or product can be used for new applications, e.g., because the copper content has been reduced. The end product can be a prime aluminum product from IBA, which can be highly useful, e.g., in sheet alloy applications. It can have very good corrosion-resistance, finish-ability and strength. Specific embodiments can provide an aluminum-alloy clad sheet having high strength and excellent erosion resistance.

[0033] Although specific embodiments of the disclosure have been described above in detail, the description is merely for purposes of illustration. It is to be understood that the present description illustrates those aspects of the invention relevant to a clear understanding of the invention. Certain aspects of the invention that would be apparent to those of ordinary skill in the art and that, therefore, would not facilitate a better understanding of the invention have not been presented in order to simplify the present description. Although embodiments of the present invention have been described, one of ordinary skill in the art will, upon considering the foregoing description, recognize that many modifications and variations of the invention may be employed. All such variations and modifications of the invention are intended to be covered by the foregoing description and the following claims.

Claims

1. A method for recovering purified aluminum from a waste material, comprising the steps of: a. providing a waste material comprising aluminum and heavy metals; b. rough processing the waste material to remove heavy metals; c. floating the waste material to remove additional impurities; d. sizing the waste material into predetermined size ranges e. comminuting the waste material using a ball mill; f. wet screening the comminuted waste material to classify oversize particles; g. washing the waste material with an acid to produce darkened non-aluminum material; h. sorting the waste material to separate the darkened non-aluminum material from the purified aluminum material; and i. purifying the purified aluminum material using an eddy current separator, wherein the purified aluminum material has an aluminum purity of at least 98%.
2. The method of claim 1, wherein the rough processing step (b) includes the use of an eddy current separator to remove ferrous metals from the waste material.
3. The method of claim 1, wherein the acid used in the washing step (g) is sulfuric acid with a concentration of approximately 40%.
4. The method of claim 1, wherein the sorting step (h) includes the use of at least one optical sorter to distinguish between darkened non-aluminum materials and gray aluminum based on color differences.

5. The method of claim 1, wherein the predetermined size ranges in the sizing step (d) include particles having dimensions of less than 2.0 mm, between 2.0 mm and 6.0 mm, between 6.0 mm and 18.0 mm, and greater than 18.0 mm.
6. The method of claim 1, wherein the comminuting step (e) involves using a ball mill with balls having a radius between 4 mm and 5 mm to pulverize non-metallic attachments and flatten metallic particles.
7. A method for separating waste material, comprising the steps of: a. providing the waste material in a slurry having aluminum and heavy metals, b. rough processing the waste material to remove the heavy metals, wherein the rough processing includes density separation, c. comminuting the material, wherein the materials are removed after the heavy metals are removed through rough processing, d. floating the waste material to remove additional impurities; e. wet screening the material, f. washing the material with an acid to produce darkened material, and g. sorting the waste material to separate the darkened non-aluminum material from the purified aluminum material.
8. The method of claim 7, comprising separating the purified aluminum product using an eddy current.
9. The method of claim 7, wherein the rough processing includes density separation at or about 3.5 SG.
10. The method of claim 7, wherein the comminution includes a ball mill.
11. The method of claim 7, wherein the waste material is incinerator bottom ash zorba.
12. The method of claim 7, wherein the purified aluminum is recovered at efficiencies of greater than about 95%.
13. The method of claim 7, wherein the purified aluminum product contains greater than 97 percent by weight of aluminum.
14. A system for recovering purified aluminum from a waste material comprising: a. a source of waste material containing aluminum and heavy metals; b. a rough processing device configured to remove heavy metals from the waste material; c. a flotation device configured to remove additional impurities from the waste material; d. a sizing device configured to classify the waste material into predetermined size ranges; e. a comminution device comprising a ball mill configured to pulverize and flatten the waste material; f. a wet screening unit configured to classify oversize particles from the comminuted waste material; g. an acid washing unit configured to wash the waste material with an acid to produce darkened non-aluminum material; h. a sorting unit configured to separate the darkened non-aluminum material from the purified aluminum material; and i. an eddy current separator configured to further purify the purified aluminum material, wherein the system produces a purified aluminum material with an aluminum purity of at least 98%.
15. The system of claim 1, wherein the sorting unit includes at least one optical sorter configured to distinguish between darkened non-aluminum materials and gray aluminum based on color differences.
16. The system of claim 1, wherein the comminution device comprises a ball mill with balls having a radius between 4 mm and 5 mm, configured to pulverize non-metallic attachments and flatten metallic particles.
17. A system for preparing an aluminum product, comprising: a. a source of waste material in a slurry having aluminum and heavy metals; b. a rough processing device capable of removing the heavy metals from the waste material, wherein the rough processing includes density separation; c. a flotation device configured to remove additional impurities from the waste material d. a comminution device that is a ball mill and that pulverizes the material; e. a screen; f. an acid washing unit configured to wash the waste material with an acid to produce darkened non-aluminum material; g. a sorting unit configured to separate the darkened non-aluminum material from the purified aluminum material; and h. a collector for the aluminum product.
18. The system of claim 17, wherein the acid is sulfuric acid.

19. The system of claim 17, wherein the rough processing device comprises an eddy current separator configured to remove ferrous metals from the waste material.

20. The system of claim 17, wherein the waste material is automobile shredder residue.
